

# Smart City Traffic Intelligence — Data Analytics Insights Report

*Scope: 48,204 hourly records of westbound I-94 traffic near Minneapolis–St Paul (Oct 2012 – Sep 2018), analysed with SQL, descriptive statistics, probability analysis and a Power BI dashboard.*

## 1. Key findings

**1.1 Demand is stable — coverage is not.** Raw yearly totals swing wildly (8.2M vehicles in 2012 to 35.4M in 2017), but this is driven by sensor coverage: hours recorded range from 2,559 (2012) to 10,605 (2017). Normalised, average hourly volume is stable at 3,169–3,341 vehicles/hour across all six years. The apparent "+20.1% growth in 2017" is better data collection, not more traffic. New Year's Day 2015 is entirely missing — a coverage gap, not zero traffic.

**1.2 Traffic is a peak-driven system.** Mean hourly volume is 3,260 vehicles but the standard deviation is ~1,987 (61% of the mean); half of all hours fall between 1,193 and 4,933. In 2017, the twin peaks are 07:00 (~4,880) and 16:00 (~5,798) versus a 03:00 trough (~382). Congestion (>5,500 vehicles/hour) occurs in **14.7% of all hours** — almost entirely rush hours.

**1.3 Weather matters far less than the calendar.** The temperature–traffic correlation is weak (Pearson  $r = 0.13$ , explaining 1.7% of variance). Clouds show the highest average traffic (3,618/h) and fog the lowest among well-sampled conditions (2,704/h) — a 915/h gap — but squall's apparent minimum (2,062/h) rests on only 4 records and is not statistically meaningful. Probability analysis confirms it:  $P(\text{Congestion}) = 14.7\%$ , and the odds of congestion in clear weather are  $0.74\times$  those in cloudy weather — weather and congestion are not independent, but the relationship is mild and secondary to time-of-day.

**Table 1** Average hourly traffic by weather condition (well-sampled conditions highlighted)

| Weather condition                | Avg traffic (veh/h) | Records |
|----------------------------------|---------------------|---------|
| <b>Clouds — highest</b>          | <b>3,618</b>        | 15,164  |
| Haze                             | 3,502               | 1,360   |
| Rain                             | 3,318               | 5,672   |
| Clear                            | 3,056               | 13,391  |
| Snow                             | 3,017               | 2,876   |
| Mist                             | 2,933               | 5,950   |
| <b>Fog — lowest well-sampled</b> | <b>2,704</b>        | 912     |
| Squall — lowest (only 4 records) | 2,062               | 4       |

**1.4 Holidays suppress traffic by ~20%.** Holidays average 67,564 vehicles/day versus 84,978 on non-holidays. Holiday temperature differences year-on-year (e.g., New Year's Day  $-6.1^\circ\text{C}$  in 2016 vs  $-3.1^\circ\text{C}$  in 2017) do not explain this — the holiday effect dominates any weather effect.

## 2. Implications for the smart-city mobility team

- **Target the peaks, not the average:** congestion management resources (signal timing, incident response) should concentrate on 06:00–09:00 and 15:00–18:00 on weekdays.
- **Weather-based interventions are low priority:** demand management and scheduling will outperform weather-responsive measures, because weather explains little of the variance.
- **Fix observability before prediction:** partial-year coverage, a missing New Year's Day 2015, 7,629 duplicate timestamps, literal "None" holiday strings and 0-Kelvin / 0-vehicle sensor artifacts must be handled in the data pipeline (Part 2) before any ML modelling (Part 3).

## 3. Limitations

Single corridor (one sensor location) — findings do not generalise city-wide; no incident/accident data; sensor coverage varies by year; the holiday flag marks only the 00:00 hour of each holiday.